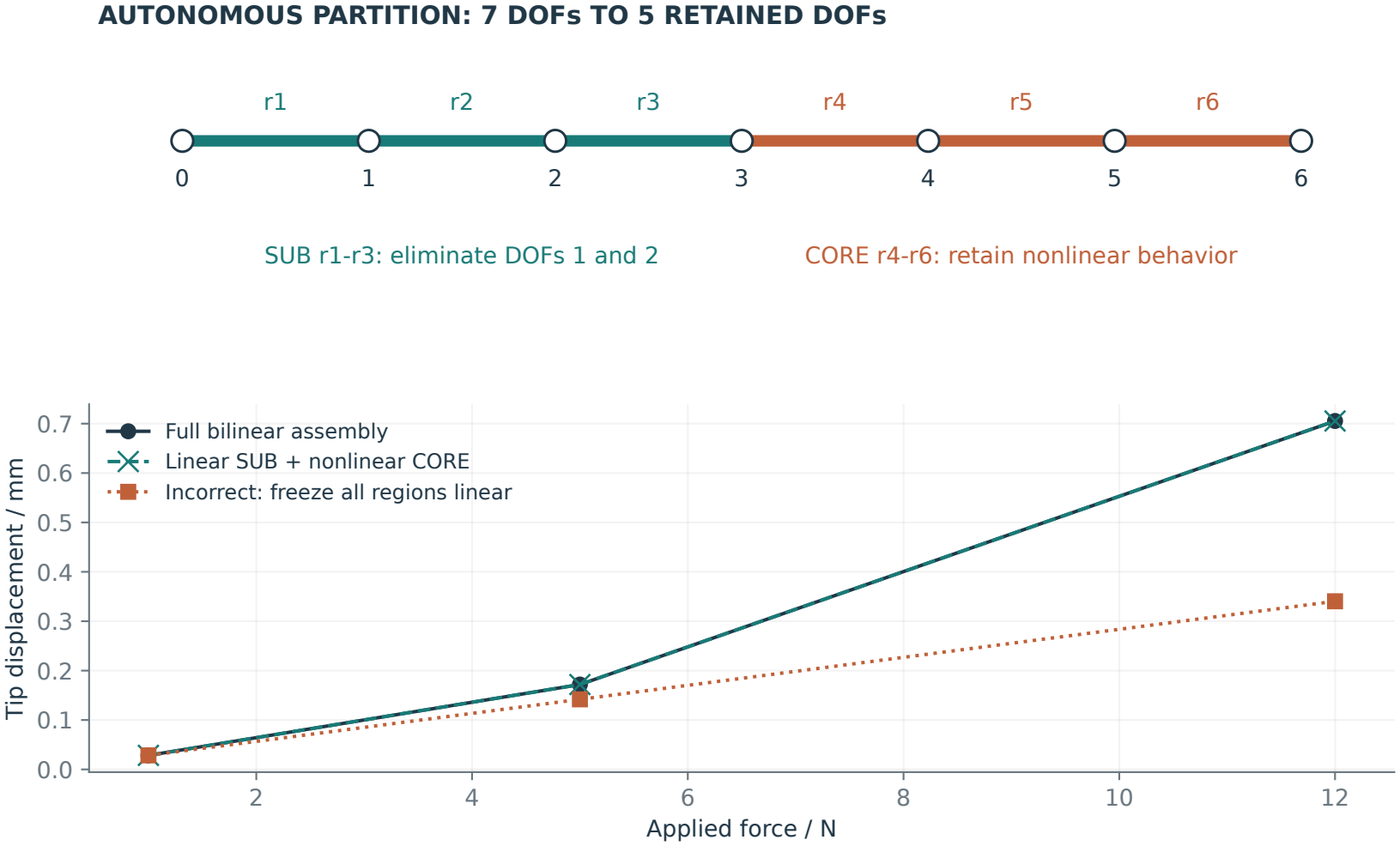
AUTONOMOUS PARTITION: 7 DOFs TO 5 RETAINED DOFs
r1
r2
r3
r4
r5
r6
0
1
2
3
4
5
6
SUB r1-r3: eliminate DOFs 1 and 2
CORE r4-r6: retain nonlinear behavior
Full bilinear assembly
Linear SUB + nonlinear CORE
Incorrect: freeze all regions linear
Tip displacement / mm
Applied force / N
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
2
4
6
8
10
12